TRAVELXP

C323 CAPSTONE

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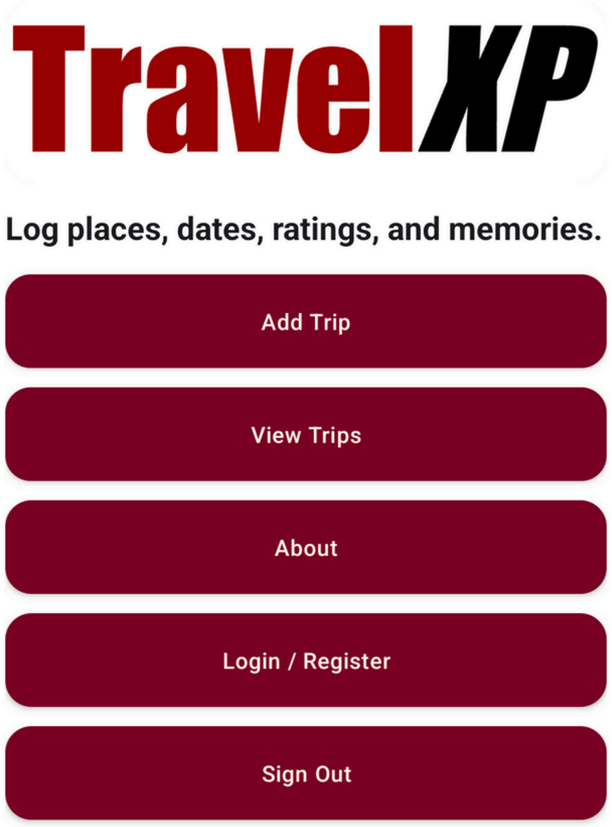
A Simple, Yet Effective Way to Track Your Travel

- Lets users record, view, update, and delete trips
- Trip data is stored in Firebase Firestore
- Multiple Android components are used together (Activities, RecyclerView, dialogs, SharedPreferences) to fulfill the architecture requirements while supporting the app’s workflow

Features

- Users can add trips with fields like name, dates, notes, and rating using multiple widget types (Spinner, DatePicker, etc.)
- A list of trips is displayed using a RecyclerView to manage dynamic data and handle user interaction
- Tapping a trip opens a detail screen where users can edit or delete entries
- SharedPreferences to save draft entries and remember filter choices, enforcing persistent local state beyond Firestore



Implementation Timeline

- Week 1 – Setup + Core Add Flow: Project + Firebase setup, navigation across activities, Trip Form with required widgets, validation, Firestore create, and SharedPreferences draft saving via lifecycle (regulated by Sarvesh)
- Week 2 – Trip List + Detail + Full CRUD: RecyclerView trip list from Firestore, detail screen, intent-based navigation, edit (prefill + update), and delete with AlertDialog (regulated by Akshath)
- Week 3 – Filters + Menu + Polish: Filtering/sorting + persisted filter choice, menu navigation to About screen, lifecycle logging for demo, final UI polish/testing + presentation prep (regulated by both)

Activities

Trip List

Trip Form

Trip Detail

About

Your Trips

Newest Oldest

asdfasdf 5.0
asdfasdf 5.0
Dec 30, 2025 → Dec 31, 2025
★★★★☆

Tokyo
Tokyo
Dec 26, 2025 → Dec 31, 2025
★★★★☆

asdfasdf 4.0
asdfasdf 4.0
Dec 15, 2025 → Dec 18, 2025
★★★★☆

asdfasdf 3.0
asdfasdf 3.0
Dec 15, 2025 → Dec 16, 2025
★★★★☆

+

Add Trip

Tokyo

Rating

★★★★☆

Start: Dec 26, 2025

End: Dec 31, 2025

Sarvesh

Visited many districts and monuments

Save

Clear Draft

≡

Tokyo

Dates: Dec 26, 2025 - Dec 31, 2025

Rating: 4/5

With: Sarvesh

Visited many districts and monuments

Edit Delete

TravelXP is a personal travel journal.

You can add trips with dates, rating, notes, and who you traveled with.

Course features shown:
- RecyclerView list
- DatePicker
- Snackbar + AlertDialog
- SharedPreferences draft saving
- Firebase Firestore CRUD

Design Choices and Development

- Architecture: Firestore backed TripRepository for CRUD; Activities coordinate UI events, data passing (tripId), and persistence using SharedPreferences for form drafts + filter restore
- Features: Multi-activity flow (Main/List/Form/Detail/About), RecyclerView + custom row layout, DatePicker/Spinner/RadioGroup/Checkbox widgets, and AlertDialog for delete confirmation
- UI choices: IU Hoosiers cream + crimson color scheme with a modern, clean look, rounded rectangular buttons, consistent spacing, and image enhanced screens for a futuristic feel

Testing

- Functional & UI Testing: Manually tested all user flows (login, add/edit/delete trips, list navigation, filters, and menu actions) to verify correct screen transitions, Firestore updates, and consistent UI behavior across activities
- Lifecycle & Data Persistence Testing: Verified SharedPreferences and lifecycle handling by rotating the device, reopening it to ensure form drafts, filter selections, and authenticated state were correctly restored without data loss

Thank You